

Partition Function and Thermodynamics of a Spin- $\frac{1}{2}$ System

D. Tong, Statistical Physics, Problem Sheet 1, Question 3.

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Consider a system consisting of N spin- $\frac{1}{2}$ particles, each of which can be in one of two quantum states, ‘up’ and ‘down’. In a magnetic field B , the energy of a spin in the up/down state is $\pm\mu B/2$ where μ is the magnetic moment. Show that the partition function is

$$Z = 2^N \cosh^N \left(\frac{\beta\mu B}{2} \right). \quad (1)$$

Find the average energy E and entropy S . Check that your results for both quantities make sense at $T = 0$ and $T \rightarrow \infty$.

Compute the *magnetisation* of the system, defined by $M = N_{\uparrow} - N_{\downarrow}$, where $N_{\uparrow/\downarrow}$ are the number of up/down spins. The magnetic susceptibility is defined as $\chi \equiv \partial M / \partial B$. Derive *Curie's Law*, which states that at high temperatures $\chi \sim 1/T$.

Solution First, consider the partition function of a system with a single particle, z :

$$z = e^{-\beta\mu B/2} + e^{+\beta\mu B/2} = 2 \cosh \left(\frac{\beta\mu B}{2} \right). \quad (2)$$

Assuming the particles are distinguishable (localised on a lattice), the partition function of a N particle system is, therefore,

$$Z = z^N = 2^N \cosh^N \left(\frac{\beta\mu B}{2} \right). \quad (3)$$

The average energy is given by

$$E = -\frac{\partial \ln Z}{\partial \beta} = -\frac{\mu B N}{2} \tanh \left(\frac{\beta\mu B}{2} \right), \quad (4)$$

and the entropy is

$$S = k_B (\ln Z + \beta E) = N k_B \left[\ln 2 + \ln \cosh \left(\frac{\beta\mu B}{2} \right) - \frac{\beta\mu B}{2} \tanh \left(\frac{\beta\mu B}{2} \right) \right]. \quad (5)$$

At $T = 0$, $\beta \rightarrow \infty$: $\ln \cosh(\beta\mu B/2) \approx \beta\mu B/2$ and $\tanh \rightarrow 1$, so the last two terms cancel and $S \rightarrow 0$, consistent with a unique ground state. For $T \rightarrow \infty$, $\beta \rightarrow 0^+$: $\ln \cosh \rightarrow 0$ and $\tanh \rightarrow 0$, giving

$$S \rightarrow N k_B \ln 2, \quad (6)$$

the maximum entropy corresponding to all 2^N microstates being equally probable.

To compute the magnetisation, let us go back to consider the single particle system. The average magnetisation of such a system is given by

$$\langle M \rangle = \frac{1}{z} (e^{-\beta\mu B/2} - e^{+\beta\mu B/2}) = -\tanh \left(\frac{\beta\mu B}{2} \right). \quad (7)$$

The magnetisation of a N -particle system is simply

$$M = -N \tanh \left(\frac{\beta\mu B}{2} \right). \quad (8)$$

For large temperature, $\beta \rightarrow 0^+$, which gives

$$\chi \approx -\frac{N\beta\mu}{2} \sim \frac{1}{T}. \quad (9)$$